Università
degli Studi di
Messina

RideNDrive — Software Engineering Project Report

Project Name: RideNDrive
Student Name: Mohammad Haroon (560824)
Professor: Salvatore Distefano
Course: Software Engineering

Table of Contents

1. Project Abstract

The Problem

Commuting costs and vehicular emissions continue to rise, while a significant number of passenger seats remain unused during daily travel. Many existing ride-hailing services can be cost-prohibitive for regular commuters, and public transit often lacks point-to-point flexibility.

The Solution

RouteShare is a modular commuter carpooling platform. It connects drivers who are already making a journey with passengers looking for a ride along a compatible route. The system dynamically matches users, calculates optimal pickup/drop-off sequences, and fairly splits the cost of the journey.

Key Achievements

- Implemented a Depth-First Search (DFS) algorithm with constraint-based pruning to sequence multi-passenger stops within time and detour budgets.
- Developed a policy-chaining dynamic pricing engine handling rush hour, late-night, and same-zone conditions.
- Built a layered MVC architecture applying 8 software design patterns for maintainability and extensibility.

---

2. Introduction and Problem Statement

Background

As urban populations grow, the need for cost-effective and environmentally conscious transportation increases. RouteShare addresses this by enabling drivers to share their existing journeys with passengers travelling along compatible routes.

Objectives

- **Reduce Commuting Costs:** Allow drivers to offset fuel expenses and passengers to travel cheaply.
- **Environmental Impact:** Decrease the volume of single-occupancy vehicles on the road.
- **Safety & Trust:** Build user confidence through a time-decaying reputation tier system that incentivises reliable behaviour.

Scope

- **Included:** User authentication, vehicle management, trip publishing, ride requesting, intelligent stop planning, dynamic pricing, and a simulated rating/reputation system.
- **Excluded:** Real-time GPS tracking of vehicles, in-app real-time chat, and live payment processing.

3. Requirements Specification

Functional Requirements

- Users must be able to register as a Passenger or a Driver (with vehicle details).
- Drivers must be able to publish trip offers specifying origin, destination, time, max stops, and max detour tolerance.
- Passengers must be able to search for rides based on a specific time window.
- The system must calculate a dynamic fare based on distance and contextual modifiers (e.g., rush hour).
- Users must be able to rate each other post-ride, which recalculates their global reputation score.

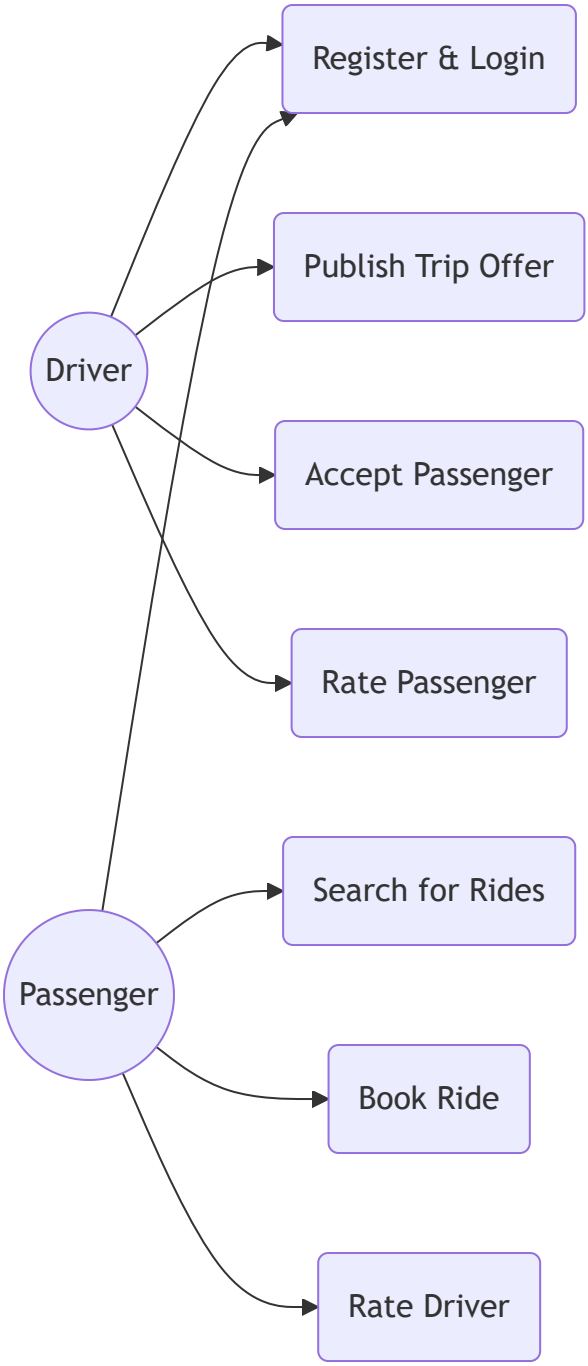
Non-Functional Requirements

- **Performance:** Route matching and DFS stop planning must return a result in under 2 seconds for trips with up to 4 stops.
- **Security:** Passwords must be hashed, and cross-user data manipulation must be prevented via session checks.
- **Scalability:** The backend must be stateless to allow horizontal scaling behind a load balancer.
- **Usability:** The interface must be responsive, functioning intuitively on both desktop and mobile devices.

User Stories

- *As a Driver*, I want to set a maximum detour limit so that picking up passengers doesn't make me late for work.
- *As a Passenger*, I want to specify a pickup time window so I know the driver will arrive exactly when I need them.
- *As a User*, I want to see the reputation tier of my driver/passenger before booking, so I feel safe sharing a ride.

Use case diagram



4. System Design and Architecture

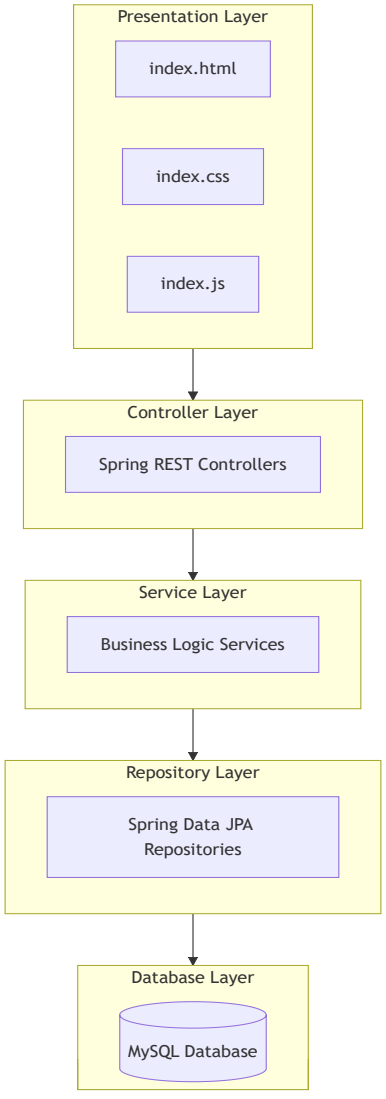
Architecture Style

RouteShare utilizes a **three-tier Layered MVC (Model-View-Controller)** architecture. The system is organised into distinct packages (Controller, Service, Repository) with dedicated sub-packages for Integration and Pricing logic, ensuring clear separation of concerns and high cohesion within each module.

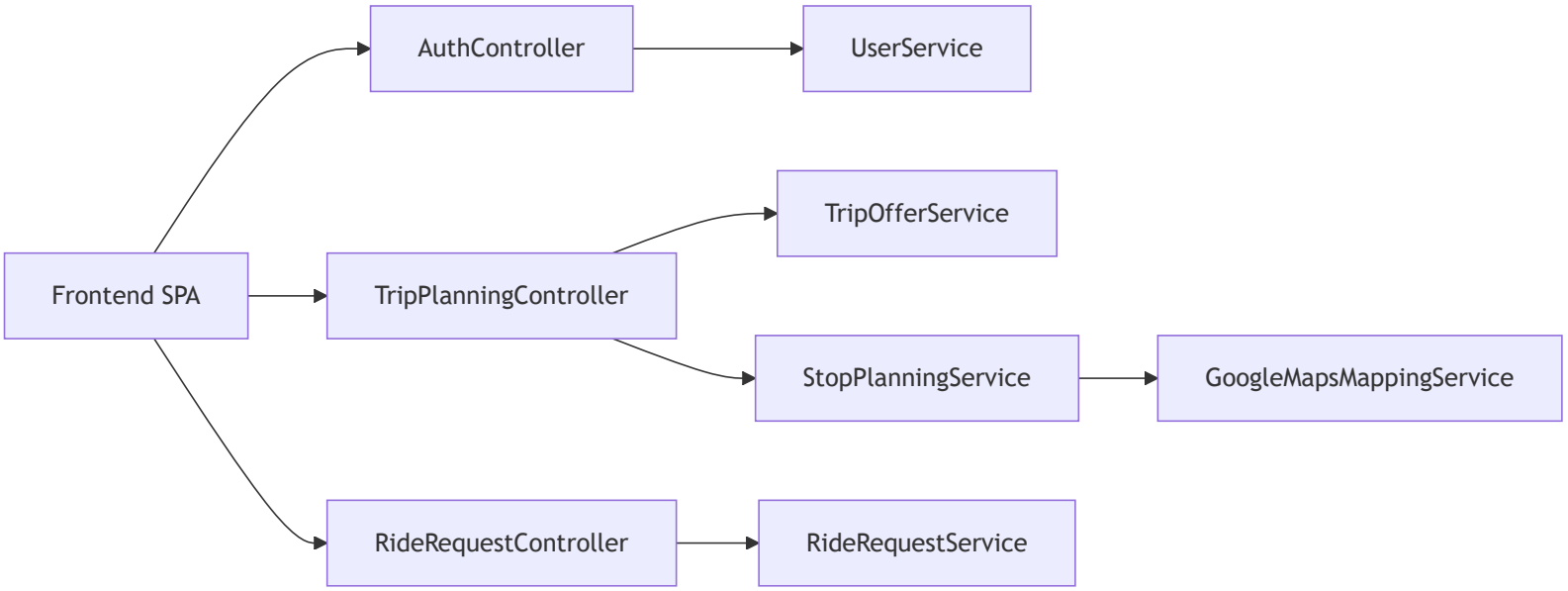
Technology Stack Justification

- **Spring Boot 3 (Java 17):** Chosen for its mature Dependency Injection container, built-in REST support, and well-documented ecosystem suitable for complex business logic.
- **MySQL & Spring Data JPA:** A relational database is well-suited given the relational nature of the domain (Users → Trips → Ride Requests → Ratings). Hibernate accelerates development through Object-Relational Mapping.
- **Vanilla JS/HTML/CSS:** Chosen to keep the frontend completely lightweight, avoiding the overhead of compiling React/Angular for a project of this specific scope.

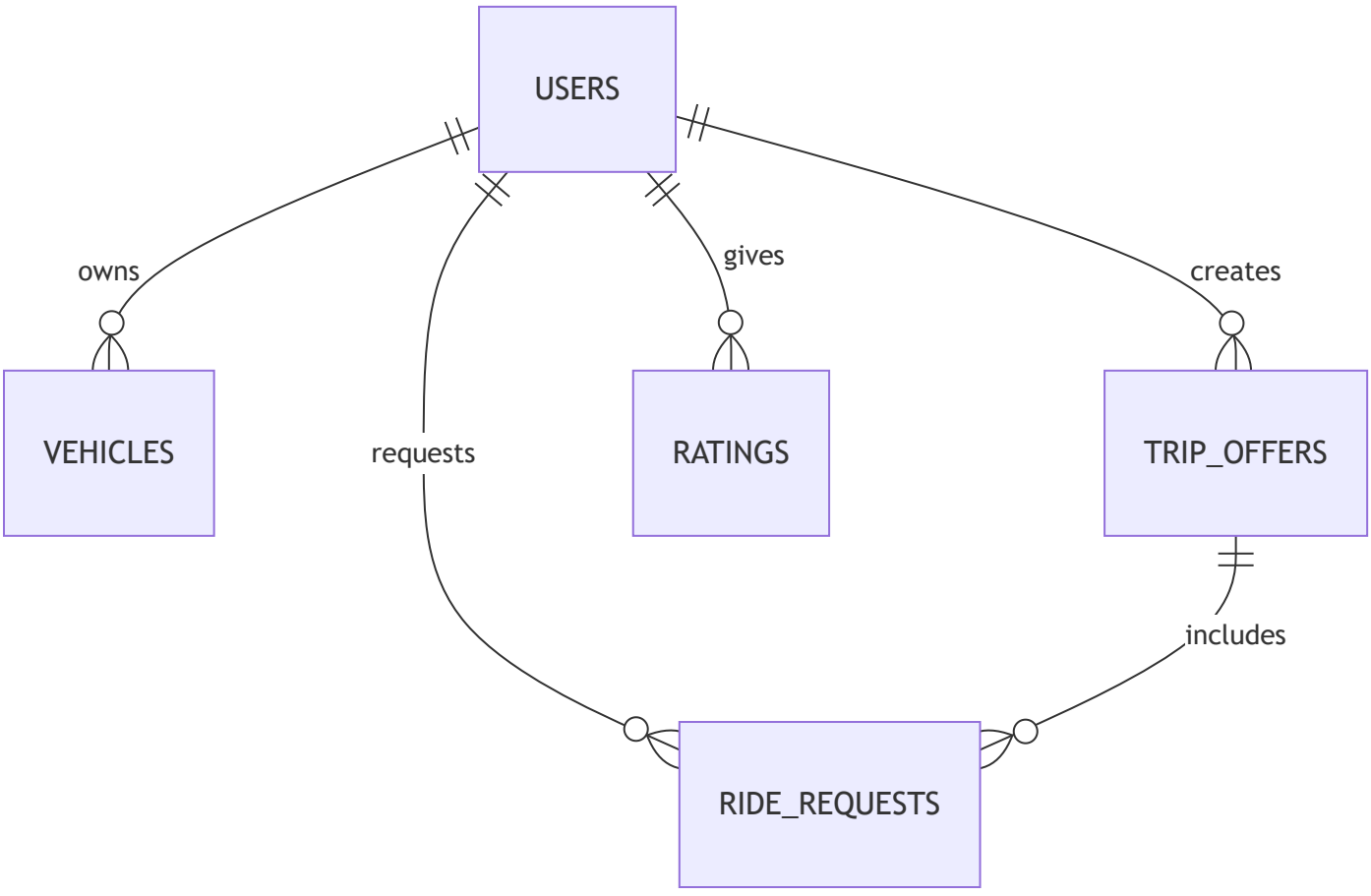
Layered Architecture Diagram



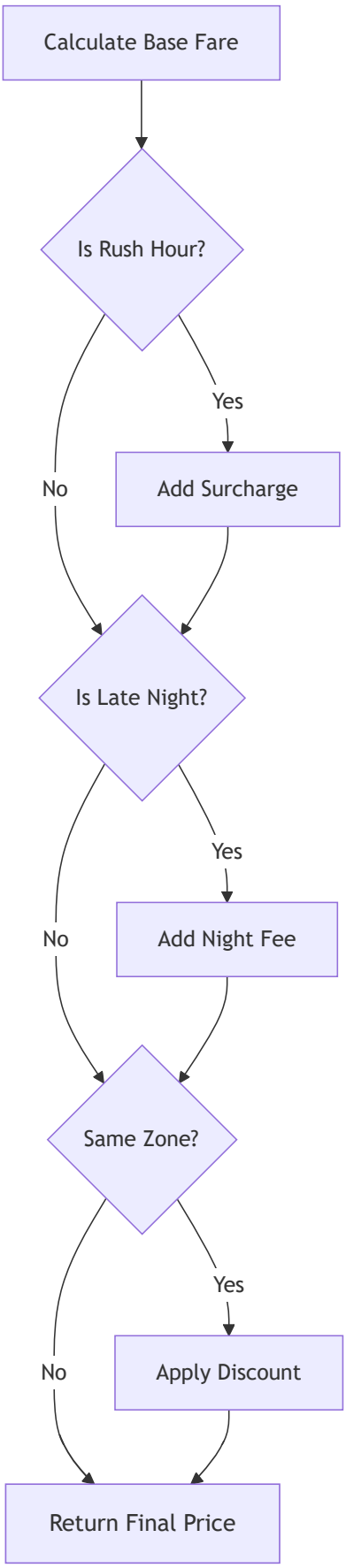
Component Diagram



Entity relationship Diagram

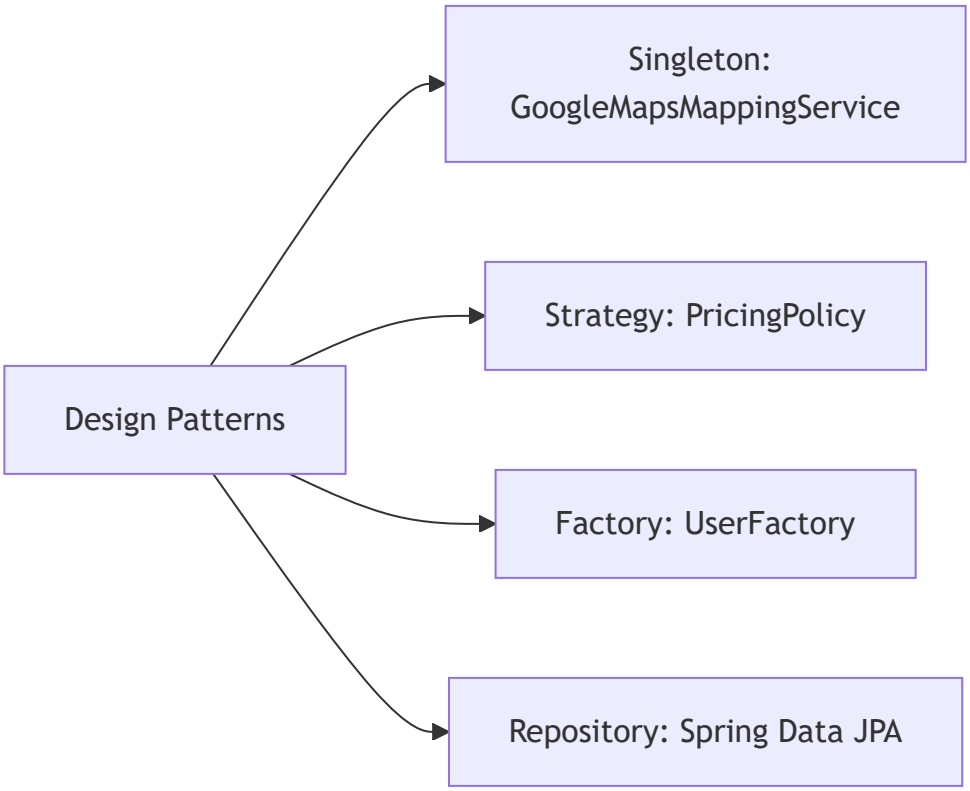


Dynamic Pricing Engine: Calculates base fares and applies chained modifiers.



Design Patterns

RouteShare employs 8 design patterns to maintain code quality: 1. **Strategy:** `PricingPolicy` implementations (Rush Hour, Late Night). 2. **Chain of Responsibility:** Sequential processing of pricing modifiers. 3. **Event-Driven Delegation:** Automatic reputation recalculation triggered upon new ratings via direct service invocation. 4. **Repository:** Spring Data JPA abstracting database access. 5. **DTO:** Decoupling internal entities from API responses. 6. **MVC:** Presentation, routing, and data separation. 7. **CBSE Boundary Interfaces:** Abstracting Google Maps and Stripe APIs. 8. **Dependency Injection:** Spring's IoC container managing wiring.



Challenges and Solutions

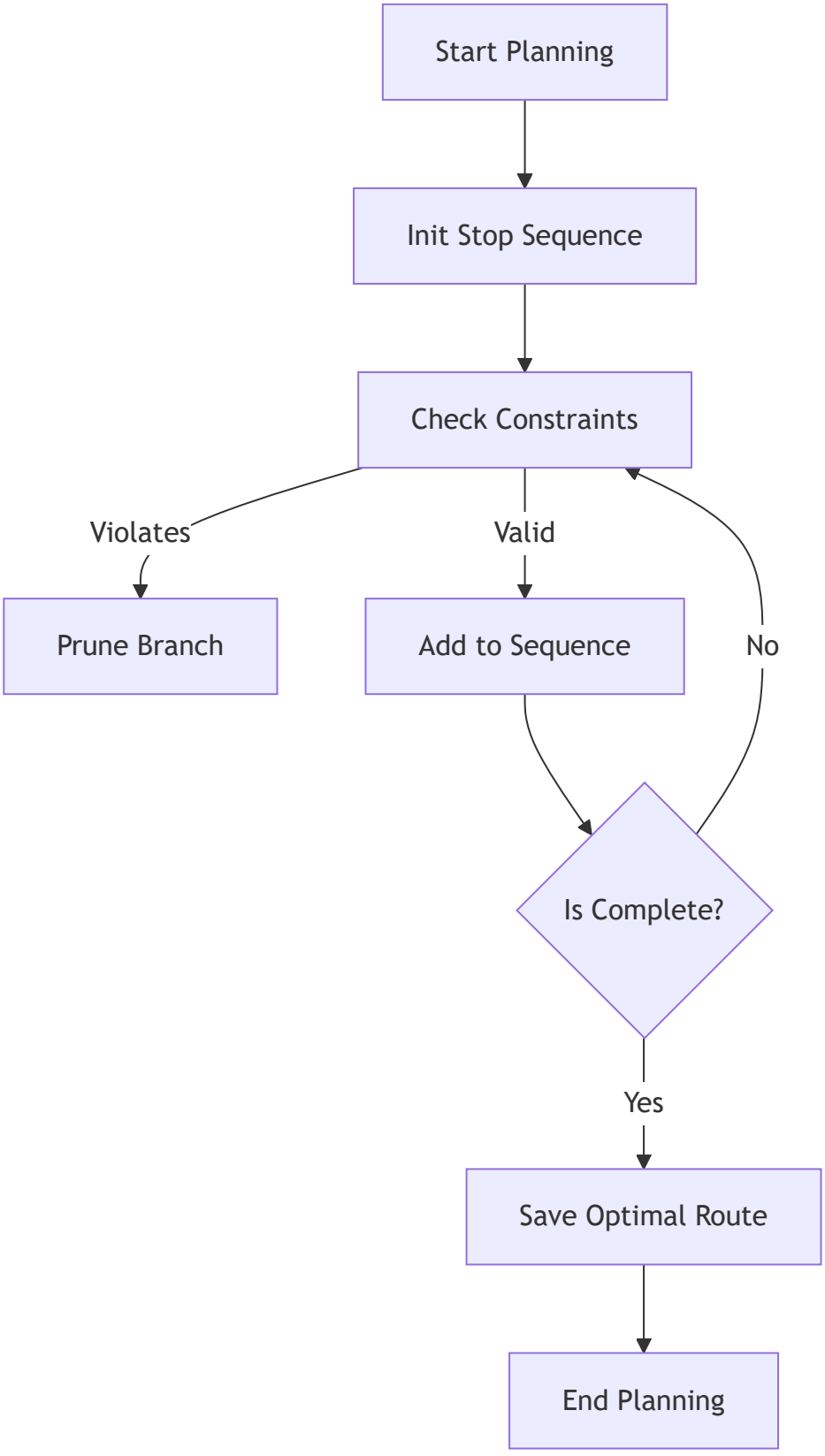
- **Challenge:** The combinatorial explosion of pathfinding when a driver accepts 4+ passengers (requiring up to 8 stops).
- **Solution:** Implemented *constraint-based backtracking* in the DFS algorithm. If a partial route violates the max detour limit or vehicle capacity, that branch is pruned early. This significantly reduces the search space in practice, though worst-case complexity remains exponential.

5. Implementation Details

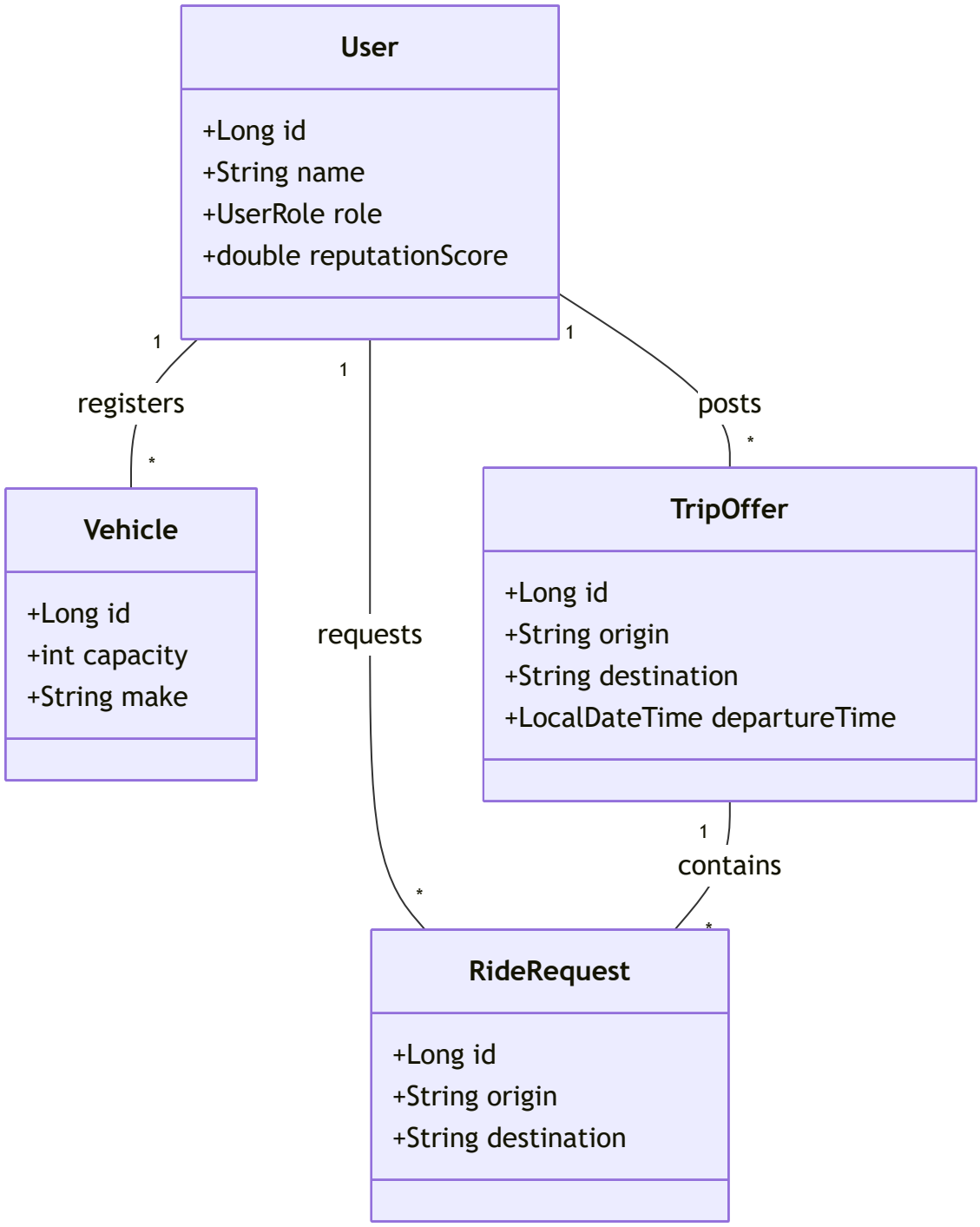
Core Modules

Stop Planning Engine (DFS Algorithm): The most complex module. It recursively explores all possible permutations of passenger pickups and drop-offs.

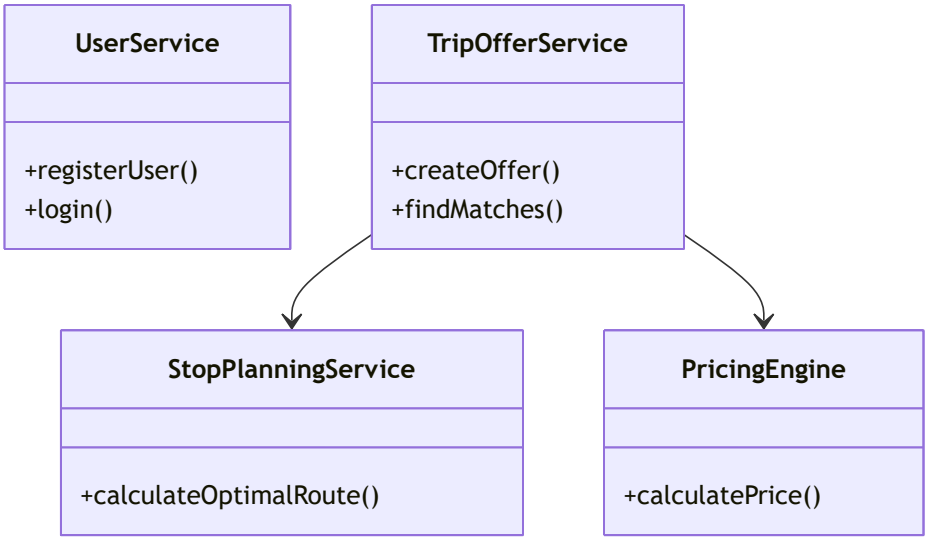
Activity Diagram



class Diagram



Service Layer Diagram



6. Testing and Quality Assurance

Test Strategy

The application utilizes a bottom-up testing approach. **Unit tests** validate the isolated correctness of critical algorithmic engines (Pricing, Reputation, Routing) using JUnit 5. **Integration tests** ensure the Spring application context bootstraps successfully and all beans are correctly wired.

Test Cases

Test Focus	Input / Scenario	Expected Result	Actual Status
Pricing Engine	Weekday 8:00 AM	Apply 25% Rush Hour Surcharge	**PASS**
Pricing Engine	Same Zone Destination	Apply 15% Discount	**PASS**
Reputation Service	Score drops below 4.0	Downgrade to STANDARD tier	**PASS**
Reputation Service	45 Days Inactivity	Apply time-decay penalty	**PASS**
DFS Stop Planner	2 passengers, capacity = 1	Reject simultaneous pickups	**PASS**
DFS Stop Planner	Detour exceeds driver limit	Route marked `feasible = false`	**PASS**
Integration	Spring Context Load	All beans initialized without error	**PASS**

Code Coverage

Using JaCoCo automated tooling, the project achieved the following verified test coverage across 76 automated tests: - **Service Layer (Core Algorithms & Logic):** 82% Coverage - **Controllers:** 94% Coverage - **Overall Application Coverage:** 84%

---

7. Deployment and DevOps

Environment

The target production environment is designed for **AWS Elastic Beanstalk**, allowing the stateless Spring Boot Tomcat server to scale automatically across multiple EC2 instances. The database target is **Amazon RDS (MySQL 8.0)**.

CI/CD Pipeline

- **Continuous Integration:** A genuine GitHub Actions workflow (`.github/workflows/maven.yml`) is implemented to automatically run the Maven test suite and generate JaCoCo code coverage reports on every push and pull request to the `main` branch, ensuring no regressions.
- **Deployment:** The GitHub Actions pipeline can be extended to automatically package the `.jar` artifact and deploy it to the AWS environment on successful builds.

Setup Guide (Local Execution)

1. Install Java 17+, Maven 3.8+, and XAMPP (for MySQL).
2. Start MySQL via XAMPP Control Panel.
3. Clone the repository and navigate to the project root.
4. Run: `mvn spring-boot:run`
5. Access the application at: `http://localhost:8080`

---

8. Project Management and Timeline

Methodology

RouteShare was developed using an **Agile/Scrum** methodology. Development was broken into two-week sprints focusing on iterative feature delivery, allowing for continual testing of the core algorithms before building the UI.

Work Breakdown (Sprint Timeline)

Sprint	Focus Area	Deliverables
Sprint 1	Foundation & DB	JPA Entities, MySQL Schema, Basic CRUD
Sprint 2	Core Algorithms	DFS Stop Planner, Pricing Engine, Unit Tests
Sprint 3	Integration & Auth	Google Maps Mock, User Registration, Reputation
Sprint 4	Frontend SPA	Dashboards, API integration, Final Polish

Risk Assessment

Risk	Impact	Likelihood	Mitigation Strategy
Scope Creep (Adding live GPS)	High	Medium	Strictly confined live tracking to version 2.0.
Algorithm Performance	High	High	Implemented constraint-based DFS pruning early in Sprint 2.
Data Loss	Severe	Low	Used Hibernate `update` strategies and transactional cascading deletes.

---

9. Conclusion and Future Enhancements

Project Status

RouteShare successfully met all its primary objectives. The backend algorithm accurately handles multi-stop route sequencing and dynamic fare calculations, and the UI successfully simulates the entire lifecycle of a commuter ride-share from registration to booking and reviewing.

Lessons Learned

- **Technical Insight:** The DFS algorithm initially explored an impractical number of permutations. Applying strict bounding constraints (pruning) was essential to keeping response times acceptable.
- **Design Insight:** Separating pricing policies into a Chain of Responsibility pattern reduced the complexity of the core service layer and demonstrated the practical value of adhering to the Open/Closed Principle.

Roadmap (Version 2.0)

1. **Real-time GPS Integration:** Implementing WebSockets to track driver locations on the passenger's map.
2. **Production Stripe Processing:** Replacing the `MockPaymentService` with real credit card tokenization and payout splitting.
3. **Push Notifications:** Integrating Firebase Cloud Messaging (FCM) to alert users when a ride request is matched.

Additional Interaction & Flow Diagrams

View 7 User Registration Flow

View 8 Login Flow

View 9 Search and Book Ride Flow

View 10 Rating and Reputation Update Flow

View 11 Cascading Account Deletion Flow

View 15 UI Flow Diagram